

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : fixed
i Dairy : categorical
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 2 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 3 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 52 events
Fitting .imp = 6 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 7 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 8 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 9 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 52 events
Fitting .imp = 11 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 12 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 13 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 14 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 50 events

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 17 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 18 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 19 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 20 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

```
-- Events Per Variable (EPV) [mice (all imputations)]
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```
i Events : 49
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```
i Coefs : 10
```

```
i EPV : 4.9
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-- C-index (Harrell's concordance) [mice (averaged)]
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i C-index : 0.794 (95% CI: 0.734-0.855)
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i SE : 0.0308
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??? C-index >= 0.70 - good discrimination.
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```
-- Global model tests [mice (averaged)]
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```
i Likelihood ratio test :  $\chi^2(10) = 65.66$ ,  $p = 4.48e-10$  <- preferred
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i Wald test :  $\chi^2(10) = 48.74$ ,  $p = 5.54e-07$ 
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i Score (log-rank) test :  $\chi^2(10) = 50.29$ ,  $p = 2.93e-07$ 
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i Total person-years (mean across imputations): NaN
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```
-- Assumption checks [mice_imp1] --
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```
-- 1. Proportional Hazards - cox.zph() | pass: global  $p > 0.05$ 
```

```
chisq df p dairy_exposure 2.10046 3 0.552 BMI 0.00146 1 0.970 education_level
```

```
5.14025 2 0.077 smoking_status 5.03219 2 0.081 mvpa_min_day_f1 0.36505 1 0.546
```

```
diabetes_status 1.70716 1 0.191 GLOBAL 15.73355 10 0.108
```

```
??? PH assumption not violated (global  $p = 0.1075$ )
```

```
Schoenfeld plots saved to
```

```
03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238
```

```
-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear
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```
'geom_smooth()' using formula = 'y ~ x'
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
Linearity plots saved to
```

-- 7. Log-log survival plot | pass: parallel lines across log(time)

Log-log plot saved to

03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238

-- 8. Cox-predicted vs KM - calibration by dairy quartile

Cox vs KM calibration plot saved to

03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238

-- Kaplan-Meier plot by dairy category [mice_imp1]

Ignoring unknown labels:

o colour : "Dairy quartile"

i Log-rank p-value: 0.596

KM plot saved to 03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high
1 Q1	271					
10.00	19203.20	0.52	0.25	0.96	2 Q2	291
10.70	20614.22	0.52	0.26	0.94	3 Q3	272
12.00	19292.60	0.62	0.32	1.09	4 Q4	287
17.45	20473.00	0.85	0.50	1.36		

Incidence table saved to

03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238

Model results saved to

03_outputs/Cox/mice_ewgsop2_fixed_categorical_27072026_1238/mice_model_results.csv